

Goal-Oriented Space–Time Adaptivity for the Heat Equation

1 PDE

Let

$$\Omega = (0, 1)^2, \quad Q_T = \Omega \times (0, T).$$

We consider the parabolic problem

$$\partial_t u - \Delta u = f \quad \text{in } Q_T, \tag{1}$$

$$u = 0 \quad \text{on } \partial\Omega \times (0, T), \tag{2}$$

$$u(\cdot, 0) = u_0 \quad \text{in } \Omega. \tag{3}$$

The manufactured exact solution is

$$u(x, y, t) = e^{-t} \sin(\pi x) \sin(\pi y). \tag{4}$$

The corresponding source term is

$$f(x, y, t) = (2\pi^2 - 1)e^{-t} \sin(\pi x) \sin(\pi y). \tag{5}$$

2 Quantity of interest

The goal functional is the spatial average over the box

$$\omega = [a, b] \times [a, b], \quad a = 0.25, \quad b = 0.75.$$

Define the indicator

$$\chi_\omega(x, y) = \begin{cases} 1, & (x, y) \in \omega, \\ 0, & \text{otherwise.} \end{cases}$$

The quantity of interest is

$$J(u) = \int_{\Omega} \chi_\omega(x, y) u(x, y, T) dx. \tag{6}$$

For the exact solution,

$$J(u) = e^{-T} \left(\frac{\cos(\pi a) - \cos(\pi b)}{\pi} \right)^2. \tag{7}$$

3 Primal and dual problems

3.1 Primal problem

The primal weak problem is: find $u(t) \in H_0^1(\Omega)$ such that

$$(\partial_t u, v) + (\nabla u, \nabla v) = (f, v) \quad \forall v \in H_0^1(\Omega). \quad (8)$$

With the time grid $0 = t_0 < t_1 < \dots < t_N = T$, the code uses a $dG(0)$ -in-time step:

$$(u_n, v) + k_n \kappa (\nabla u_n, \nabla v) = (u_{n-1}, v) + k_n (f_n, v), \quad (9)$$

where $k_n = t_n - t_{n-1}$ and $f_n = f(\cdot, t_n)$.

3.2 Dual problem

The dual problem associated with the goal functional is posed backward in time:

$$-\partial_t z - \kappa \Delta z = \chi_\omega \quad \text{in } Q_T, \quad (10)$$

$$z = 0 \quad \text{on } \partial\Omega \times (0, T), \quad (11)$$

$$z(\cdot, T) = \chi_\omega \quad \text{in } \Omega. \quad (12)$$

4 Global DWR estimator

Let U_n be the primal numerical solution at time t_n , and let Z_n be the enriched dual solution. Denote by $\Pi_h Z_n$ its projection/interpolation into the primal space and define

$$z_n^e = Z_n - \Pi_h Z_n.$$

Then the signed global estimator can be written in residual-weight form as

$$\eta_{\text{global}} = \sum_{n=1}^N \rho_n(U_n)(z_n^e), \quad (13)$$

where the discrete residual functional is

$$\rho_n(U_n)(v) := \int_{\Omega} \left(k_n f_n v - (U_n - U_{n-1})v - k_n \kappa \nabla U_n \cdot \nabla v \right) dx. \quad (14)$$

This quantity is compared against the true goal error

$$J(u(T)) - J(U_N).$$

5 Localisation

For local marking and refinement, the estimator is split into space and time parts.

5.1 Spatial residual

The cell residual is

$$R_n = f_n + \Delta U_n - \frac{U_n - U_{n-1}}{k_n}. \quad (15)$$

The facet residual is

$$r_n = \frac{1}{2} [\nabla U_n] \cdot n. \quad (16)$$

The local spatial indicator on a cell K is built from

$$\rho_{1,K,n} = \sqrt{k_n} \|R_n\|_K + h_K^{-1/2} \sqrt{k_n} \|r_n\|_{\partial K}, \quad (17)$$

$$\omega_{1,K,n} = \sqrt{k_n} \|z_n - z_{h,n}\|_K + h_K^{1/2} \sqrt{k_n} \|z_n - z_{h,n}\|_{\partial K}, \quad (18)$$

so that

$$\eta_{K,n}^{\text{space}} = \rho_{1,K,n} \omega_{1,K,n}. \quad (19)$$

5.2 Temporal indicator

The temporal indicator uses the jump in time:

$$\rho_{2,K,n} = k_n^{-1/2} \|U_n - U_{n-1}\|_K, \quad (20)$$

$$\omega_{2,K,n} = k_n^{1/2} \|z_n - z_{h,n}\|_K, \quad (21)$$

hence

$$\eta_{K,n}^{\text{time}} = \rho_{2,K,n} \omega_{2,K,n}. \quad (22)$$

5.3 Combined local estimator

The cell-wise local indicator is

$$\eta_K^{\text{abs}} = \sum_{n=1}^N (\eta_{K,n}^{\text{space}} + \eta_{K,n}^{\text{time}}). \quad (23)$$

The global estimator used for marking is

$$\eta_h = \eta_{\text{space}} + \eta_{\text{time}}, \quad \eta_{\text{space}} = \sum_K \eta_K^{\text{space}}, \quad \eta_{\text{time}} = \sum_K \eta_K^{\text{time}}. \quad (24)$$

6 Adaptive strategy

The adaptive loop alternates between solving, estimating, and refining.

6.1 Refinement decision

The code decides whether to refine in time or space by comparing the relative sizes of the two contributions:

$$\eta_{\text{total}} = |\eta_{\text{space}}| + |\eta_{\text{time}}|. \quad (25)$$

Time refinement is triggered when

$$|\eta_{\text{time}}| > \tau_t \eta_{\text{total}},$$

and space refinement is triggered when

$$|\eta_{\text{space}}| > \tau_x \eta_{\text{total}},$$

with user parameters τ_t and τ_x .

6.2 Dörfler marking

For time slabs, the code marks slabs using bulk marking: choose a smallest set of slabs whose indicator sum reaches a fixed fraction α_t of the total time indicator.

For space, cell markers are chosen analogously with fraction α_x .

6.3 Grid update

If a time slab is marked, it is bisected by inserting its midpoint:

$$t_{n-\frac{1}{2}} = \frac{1}{2}(t_{n-1} + t_n). \quad (26)$$

If a spatial cell is marked and local refinement is available, the mesh is refined locally.

6.4 Algorithm

1. Solve the primal problem on the current space–time grid.
2. Solve the enriched dual problem backward in time.
3. Compute the goal error estimator η_h and the signed global estimator η_{global} .
4. Compare space/time contributions.
5. Refine time slabs and/or spatial cells by Dörfler marking.
6. Repeat until the maximum number of adaptive iterations is reached.

7 Results

Table 1: Summary of I_{eff} across iterations (z_{num}).

it	DOFs	N_t	I_{eff}
0	83	8	-2.6128
1	122	10	-0.7516
2	175	14	0.1948
3	244	20	0.6025
4	404	29	0.9247
5	598	44	0.9912
6	896	67	1.0184
7	1408	101	1.0317
8	2105	153	1.0073

Table 2: Summary of I_{eff} across iterations ($I_h z_{\text{enriched}}$).

it	DOFs	N_t	I_{eff}
0	83	8	-2.6128
1	122	9	-0.6955
2	150	10	0.5855
3	214	11	1.1089
4	318	13	1.2813
5	383	17	1.2865
6	488	23	1.2235
7	663	31	1.2082
8	905	44	1.1924
9	1139	64	1.1486
10	1460	95	1.1377
11	1916	142	1.1150
12	2529	213	1.0944
13	3450	321	1.0820
14	4317	484	1.0699
15	5486	733	1.0587